

Source code:

```
sbit LCD_RS at RB0_bit;  
sbit LCD_EN at RB1_bit;  
sbit LCD_D4 at RB2_bit;  
sbit LCD_D5 at RB3_bit;  
sbit LCD_D6 at RB4_bit;  
sbit LCD_D7 at RB5_bit;
```

```
sbit LCD_RS_Direction at TRISB0_bit;  
sbit LCD_EN_Direction at TRISB1_bit;  
sbit LCD_D4_Direction at TRISB2_bit;  
sbit LCD_D5_Direction at TRISB3_bit;  
sbit LCD_D6_Direction at TRISB4_bit;  
sbit LCD_D7_Direction at TRISB5_bit;
```

```
char display[16];
```

```
void main() {
```

```
    unsigned int result;  
    float volt, temp;
```

```
    TRISB = 0x00;    // LCD output  
    TRISA.F0 = 1;    // RA0 input for LM35
```

```
    ADCON1 = 0x80;    // ADC configuration
```

```
    LCD_Init();  
    LCD_Cmd(_LCD_CLEAR);  
    LCD_Cmd(_LCD_CURSOR_OFF);
```

```
    while(1) {
```

```
        // Read ADC value from AN0  
        result = ADC_Read(0);
```

```
        // Convert ADC value to voltage
```

```
volt = result * 5.0 / 1023.0;
```

```
// LM35 temperature calculation
```

```
temp = volt * 100.0;
```

```
// Display on LCD
```

```
LCD_Out(1,1,"Temp = ");
```

```
FloatToStr(temp, display);
```

```
LCD_Out(1,8,display);
```

```
LCD_Chr(1,15,223); // Degree symbol
```

```
LCD_Out_CP("C");
```

```
Delay_ms(500);
```

```
}  
}
```

Output:



